

RED PINES, NB, Canada

WMO: 716340

Lat: 47.4394N

Lon: 65.5978W

Elev: 407

StdP: 14.48

Time Zone: -4.00 (NAA)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-8.3	-19.4	1.9	-9.8	-15.3	2.4	-6.3	13.6	24.3	12.1	19.8	2.4	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	84.9	67.8	81.4	66.4	78.2	64.8	70.7	79.1	69.0	76.9	67.4	74.9	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	104.4	73.7	66.3	98.3	72.4	64.8	93.2	71.2	34.9	79.6	33.4	76.9	32.2	74.8	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.3	9.4	DB	-23.6	90.1	6.5	2.9	-28.3	92.2	-32.1	93.9	-35.8	95.6	-40.5	97.7	(4)
(5)			WB	-22.8	73.9	6.9	1.4	-27.7	74.8	-31.8	75.6	-35.6	76.4	-40.7	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	12.9	14.3	23.3	36.2	49.2	57.6	66.0	64.5	55.8	44.4	32.2	20.7		
	DBStd	20.25	11.25	10.64	10.44	7.45	7.35	6.49	5.86	4.91	7.79	7.50	8.84	10.79		
	HDD50	5197	1149	1000	831	419	106	10	0	0	32	207	535	908		
	HDD65	9335	1614	1420	1294	864	495	237	58	72	287	638	983	1373		
	CDD50	1513	0	0	2	5	80	238	496	450	207	34	1	0		
	CDD65	177	0	0	0	0	4	15	90	56	11	1	0	0		
	CDH74	1818	0	0	1	2	87	232	877	462	152	5	0	0		
	CDH80	432	0	0	0	0	17	60	238	82	35	0	0	0		
(14) Wind	WSAvg	3.9	4.4	4.5	4.8	4.6	4.0	3.4	3.1	2.8	3.1	3.7	4.1	4.0		
(15) Precipitation	PrecAvg	40.9	3.3	2.7	3.2	2.8	3.6	3.3	3.7	3.5	3.2	4.1	3.6	3.7		
	PrecMax	52.8	4.7	5.1	7.0	5.4	7.1	5.6	6.6	6.5	7.6	7.4	6.6	8.4		
	PrecMin	29.4	0.9	0.3	1.0	1.3	0.9	1.1	0.9	1.0	1.2	1.0	1.4	1.1		
	PrecStd	5.1	1.1	1.1	1.3	1.1	1.4	1.0	1.3	1.5	1.6	1.7	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	46.1	63.8	67.5	81.7	86.7	89.9	85.7	84.4	72.7	61.4	49.4		
		MCWB	48.8	42.4	49.4	50.5	63.5	65.3	70.0	69.1	67.8	61.5	60.2	48.1		
	2%	DB	39.4	38.8	47.7	58.8	75.7	80.3	86.0	82.4	78.6	65.9	54.6	43.7		
		MCWB	38.3	34.4	41.4	49.9	59.0	63.9	69.4	67.0	65.7	59.0	51.4	42.3		
	5%	DB	34.3	34.1	42.2	53.7	69.8	75.7	82.3	79.0	73.1	61.5	49.6	39.5		
		MCWB	32.8	31.0	36.0	46.1	55.2	60.2	67.1	66.1	63.7	55.8	45.6	37.9		
	10%	DB	30.8	30.5	38.1	48.8	64.5	71.4	78.9	76.0	69.8	57.3	45.0	34.5		
		MCWB	29.4	28.2	32.9	41.2	53.3	58.8	65.7	64.9	62.2	52.3	42.1	33.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	42.7	52.4	56.5	66.2	68.5	73.8	71.9	70.5	63.9	60.2	48.9		
		MCDB	48.9	45.8	61.2	61.0	76.8	78.0	81.7	79.4	79.9	67.7	61.2	49.3		
	2%	WB	38.1	35.4	43.2	51.5	62.7	66.2	71.1	70.2	67.7	61.0	53.0	42.9		
		MCDB	39.3	37.6	45.0	56.9	71.3	75.1	80.0	78.3	73.5	64.9	54.4	43.8		
	5%	WB	33.1	31.3	37.1	46.4	58.7	63.7	69.5	68.4	65.9	57.1	46.3	38.9		
		MCDB	34.0	33.5	40.7	54.0	64.7	72.1	77.7	75.7	71.8	61.9	48.5	39.8		
	10%	WB	29.4	28.5	33.3	42.3	55.2	61.3	68.1	66.7	63.4	53.2	42.1	33.8		
		MCDB	30.6	30.2	36.6	47.8	63.5	68.1	76.4	73.3	69.3	56.6	44.7	34.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	21.0	20.5	18.3	22.8	21.3	22.0	21.0	21.3	17.7	14.0	14.6		
		MCDBR	20.2	23.8	24.3	27.3	32.7	30.5	26.3	24.8	26.0	24.3	20.2	15.4		
		MCWBR	20.0	20.6	17.3	17.5	17.7	15.4	12.1	12.5	15.1	17.2	17.7	15.0		
	5% WB	MCDBR	20.7	21.0	20.2	24.2	25.0	22.7	21.4	20.4	22.4	20.2	18.2	14.8		
		MCWBR	20.8	19.1	16.1	17.8	15.6	13.3	11.3	11.5	15.3	15.9	18.1	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.276	0.311	0.364	0.363	0.382	0.412	0.387	0.355	0.326	0.295	0.277		
	taud		2.378	2.348	2.362	2.329	2.404	2.349	2.302	2.368	2.445	2.519	2.548	2.462		
	Ebn at Noon		258	285	289	282	286	279	269	272	271	264	252	246		
	Edh at Noon		22	28	32	37	36	38	40	36	30	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		405	677	1063	1367	1629	1756	1755	1535	1192	742	403	297		
	RadStd		48	77	77	116	154	132	130	97	102	62	50	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page